PERFORMANCE
CONSTRUCTION CHEMICALS

SWADI®
a step ahead...

AG02 Pump Grade

Free Flow, High Early Strength, Non - Shrink, Cementitious Precision Grout & Chloride - Free

Description

SWADI - AG02 Pump grade is supplied as a ready to use dry powder. The addition of controlled amount of clean water produces a free flowing, non-shrink grout for concrete tie rod hole/slits. AG02 Pump grade is a blend of Portland cement, graded fillers and chemical additives. The grade fillers are designed to assist uniform mixing and produce a consistent free flow grout.

Consistency in performance of grout is assured, as only recommended quantity of water is required to be added to the designed premix composition. The pumpable and self-compact properties ensure complete filling voids and perfect bearing.

Standard Compliance

Complies with Standards IS 4031 (Part 6): 1988, IS 10158: 1982, IS 4031 (Part 5): 1988, IS 516, ASTM C 531.

Where to Use

SWADI AGO2 Pump grade grouts are ideally suited to withstand static and dynamic loads. Typical applications would be the grouting for tie rod holes / slits in concrete structure, anchor bolts, void filling requirements, bore packing, Casting Cover blocks and also many other equipment foundations.

Advantages

- Mix consistency ensures high level of contact with load bearing area.
- Provides excellent dimensional stability under temperature variations.
- High strength ensures the durability of the hardened grout.
- Develops high early strength without the use of chlorides.
- Does not segregate even at high flow – while pumping.
- Versatile, can be applied in any consistency from dry pack to fluid just by varying water content.
- Mainly ease application and max field control, just add water, mix and place.

Product Data

Color	Concrete grey
Application temperature range	From 5°C to 45°C
Powder density	1470 Kg/m³
Mix density	1850 Kg/m³
Mix ratio (w/p)	250 to 260 ml/kg for Pumpable mixture
Open time	1 hour
Set time	24 hours

Technical Data

Compressive strength IS 4031(Part 6): 1988 Pumpable (w/p-0.23%)	28 days: 34 to 38 MPa
Tensile strength	1.2 N/mm² @28 Days
Flexure strength	5.1 N/mm² @28 Days
Shrinkage	Nill
Coefficient of thermal expansion	Nill $^{-6}$
Bleeding IS 9103:1999	0% bleeding
Chloride content IS 456:2000	<0.1%

AG02 Pump Grade

Application Instructions

Surface Preparation

The Substrate/Tie rod hole must be free from oil, grease or any loosely adherent material. If the concrete surface is defective or has laitance, it must be cut back to a sound base. Tie road holes and fixing pockets must be blown clean of any dirt or debris.

Mixing

- For best results a mechanically powered grout mixer should be used. When quantities up to 50kg are used, a heavy-duty slow speed drill (400-500 rpm) fitted with a paddle is suitable.
- The quantity of clean water required to be added to a 25kg bag to achieve the desired consistency (Pumpable: Water ratio).
- The selected water content should be accurately measured into the mixer; this will ensure that the grout has a smooth even pump consistency.

Application

Mixed grout may be pumped or transported by buckets or wheelbarrows without any delay, Swadi AG02 Pump Grade mortar can be applied into tie rod holes using a suitable mortar gun within 1 hour of mixing.

- Use a mortar pump or caulking gun to inject the mixed mortar directly into the holes.
- After pumping, use a spatula, putty knife, or similar tool to level and finish the surface.
- For final cleaning, gently wipe the area with a damp sponge or cloth after the initial set, ensuring a clean and smooth finish.

Coverage

Coverage per 25 kg bag of AG02 Pump Grade material:

10 mm diameter × 200 mm length: ~168 holes.
6 mm diameter × 300 mm length: ~314 holes.

Substrate conditions (e.g., roughness, porosity) may reduce coverage due to increased material use. Smooth substrates optimize coverage.

Curing

Self-Curing Mixture.

Shelf Life

SWADI - AG02 PG has a shelf life of 12 months if kept in dry store in sealed bags, if exposed to rain or high temperatures shelf life may be reduced.

Health & Safety

For further information, please consult the Material Safety Data Sheet (MSDS).

Condition of Sale

Sold subjects to the company's condition of sale which are available on request.

Packaging

25 kg bag.

Storage and Handling Guidelines

The product should be stored in a dry, well-ventilated area, away from moisture, direct sunlight, and children, and in compliance with national regulations. Ensure the packaging is tightly sealed to prevent contamination or moisture ingress, and store it within the recommended temperature range of 5°C to 35°C. Handle the product with care, avoiding direct contact with skin and eyes, and wear appropriate protective equipment such as gloves and masks to prevent inhalation of dust. Use the product as per the instructions provided on the packaging for best results.

Environmental, Health, and Safety Information

For comprehensive information regarding the safe transportation, handling, storage, and disposal of this chemical product, please refer to the accompanying Safety Data Sheet (SDS). The SDS contains detailed information about physical properties, environmental impacts, toxicological hazards, and other safety-related factors. It is crucial to read the current SDS before using any product.

Disclaimer

The information provided in this datasheet is based on our extensive experience and is given in good faith to assist you. However, we reserve the right to update this information at any time without prior notice due to our ongoing research and development efforts.

While we guarantee the consistent quality of our products, we cannot be held responsible for any losses or damages that may arise from their use if site conditions or the execution of work are not properly managed. This data sheet becomes invalid upon the publication of a new edition, and please contact us for the latest version. Please note that we do not provide any warranties regarding the results of our products in specific applications, as the storage, handling, and application of materials are factors beyond our control.

Contact Information

For immediate orders or enquires, contact our **24/7 support team at +91 8792919513 or enquiry@swadigroup.co.in.** For additional support options and resources, please visit our website at swadigroup.co.in